JOURNAL OF OPTIMIZATION THEORY AND APPLICATIONS: Vol. 69, No. 2, MAY 1991

Diffusion Equation Method of Global Minimization: Performance for Standard Test Functions[1]

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Communicated by L. C. W. Dixon

Abstract. Recently, we have proposed a method of global minimization that is based on solving the diffusion equation with the objective function as a boundary condition. In the present paper, the performance of the method is examined when applied to the standard test functions of the Goldstein-Price, Hartman, Shekel, and Griewank families. It turns out that the method succeeds in all these cases. A comparison of the effectiveness of our method and other methods is also reported.

Key Words. Global optimization, global minimum, diffusion, multiple minima.

1. Introduction

Global minimization of a multivariate function represents a field of real challenge. The importance of the problem is obvious and comes from practically all domains of science and technology. As stated by Dixon *et al.* in 1975 (Ref. 1), "the present situation with regard to solving global optimization problems is still very poor." After 13 years, despite considerable progress achieved, the problem has not been solved and one needs more than ever a reliable global optimization method.

On the other hand, it has been proved that it is impossible to construct an algorithm that would be able to locate the global minimum of any function (Ref. 2). Hopefully, the functions encountered in physics are less pathological, because there the value of a function in an infinitesimal domain

[1] This work was partly supported by the Polish Academy of Sciences within Projects CPBP 01.12 and CPBP 01.15.

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is of negligible interest, while its behavior within a large domain is important. Hence, methods extracting low and wide minima (not necessarily the lowest ones) may be accepted from a practical point of view.

Recently, we have proposed a global minimization method (Ref. 3) that is based on the solution of the diffusion equation with the objective function as a boundary condition. This method is employed here.

2. Diffusion Equation Method

2.1. Basic Idea. Let us take a function $f(x)$ having many minima. We call it the objective function, and our aim is to find its global minimum. When applying a local minimization procedure to this end, one faces the problem of guessing the proper starting point. The whole space may be divided into regions of attraction (Ref. 4), each of them containing all those starting points x for which one finally gets a given local minimum by using a gradient-trajectory algorithm. Hereafter, whenever we mention the term "depth of a region of attraction," it means the difference between the lowest f-values at the boundary of the region and within it.

Sometimes in physical applications (Ref. 5), the number of local minima is huge (it can be easily of order 10^{100}); at the same time, a vast majority of these minima is usually meaningless, since they correspond to large values of the objective function. Of course, the barriers separating the regions of attraction are impossible to be surmounted when applying a local minimization procedure. Therefore, screening all substantially different starts in such a procedure is practically impossible.

The method in the present paper is designed to be independent of the starting point used. The main idea behind the diffusion equation method (Ref. 3) is to deform $f(x)$ onto $F(x, t)$ using a deformation parameter $t > 0$, such that $F(x, 0) = f(x)$. The deformation procedure is designed in such a way that, when t increases, any region of attraction changes its size and shape and becomes shallower. Some regions of attraction, especially the shallow ones, disappear and other ones grow at their expense. Continuing the process, for a sufficiently large $t > t_0$, one hopefully gets a single region of attraction coming from the region of attraction of the global minimum of $f(x)$ that has grown at the expense of all other regions. The position of this single minimum is attainable by a gradient-trajectory algorithm from any starting point. Then, the algorithm called "reversing procedure" is applied. In the reversing procedure, one minimizes $F(x, t_0 - \Delta t)$ with respect to x using as starting point the minimum position of $F(x, t_0)$ and then iterating until $t = 0$, the value corresponding to no deformation. Hopefully, after the reversing procedure, one obtains the global minimum.

2.2. Deformation. The type of deformation used needs some comments. For the sake of completeness, the main equations of the diffusion equation method (Ref. 3) are given below.

Let us consider a differentiable function $f(x)$, where x is a variable. Then, let us consider the transformation

$$f^{[1]}(x) = f(x) + \beta f''(x), \qquad \beta > 0, \tag{1}$$

which evidently destabilizes all regions of attraction for small β, in the sense of decreasing their depth. One may repeat this transformation N times and obtain

$$f^{[N]}(x) = (1 + \beta(d^2/dx^2))^N f(x), \qquad \beta > 0. \tag{2}$$

Taking $\beta = t/N$ and letting $N \to \infty$, one gets

$$\begin{aligned} F(x, t) &= \lim_{N\to\infty} (1 + (t/N)(d^2/dx^2))^N f(x) \\ &= \exp(t(d^2/dx^2)) f(x), \end{aligned} \tag{3}$$

where the following definition is used for the operator A:

$$\exp(A) = 1 + A + (1/2)A^2 + (1/3!)A^3 + \cdots .$$

The operator

$$T(t) = \exp(t(d^2/dx^2)) \tag{4}$$

has as eigenfunctions $\sin \omega x$ and $\cos \omega x$,

$$T(t) \sin \omega x = a(\omega, t) \sin \omega x, \tag{5}$$

$$T(t) \cos \omega x = a(\omega, t) \cos \omega x, \tag{6}$$

with eigenvalue

$$a(\omega, t) = \exp(-\omega^2 t). \tag{7}$$

This means that $T(t)$, acting on f represented by its Fourier expansion, filters out mainly the high-frequency components. We note that the operator $T(t)$, acting on polynomials, does not change their degree (see the Appendix). In particular,

$$T(t)x = x, \tag{8a}$$

$$T(t)x^2 = x^2 + 2t, \tag{8b}$$

$$T(t)x^3 = x^3 + 6tx, \tag{8c}$$

$$T(t)x^4 = x^4 + 12tx^2 + 12t^2. \tag{8d}$$

It is easy to show that $F(x, t)$ satisfies the diffusion equation

$$\partial^2 F/\partial x^2 = \partial F/\partial t, \tag{9}$$

where the parameter t denotes the time.

In n dimensions, let

$$T(t) = \exp(t\Delta),$$

Δ being the Laplacian

$$\Delta = \sum_{i=1}^{n} \partial^2/\partial x_i^2. \tag{10}$$

The many-dimensional diffusion equation reads

$$\Delta F = \partial F/\partial t. \tag{11}$$

When one is interested in the solution of (11) for the boundary condition given by

$$F(x, 0) = f(x), \qquad x \in R^n,$$

one has the Fourier-Poisson formula (Ref. 6)

$$F(x, t) = (2\sqrt{\pi t})^{-n} \int_{\mathbb{R}^n} f(y) \exp(-\|y - x\|^2/4t)\, dy, \tag{12}$$

valid for any bounded f. In the examples given below, we use Eq. (12). When the series involved in (4) converges, we are also allowed to use directly the operator $T(t)$ to produce $F(x, t)$.

3. Results and Discussion

The method described above has been applied to a commonly accepted set of eight standard test functions for global optimization (Ref. 7). All the constants necessary to define the functions are in Ref. 7. Application of the method to the particular functions is given in the Appendix (Section 4).

The Goldstein-Price function (GP, Ref. 7) is a polynomial in two dimensions:

$$\begin{aligned}\mathrm{GP}(x_1, x_2) = {} & [1 + (x_1 + x_2 + 1)^2(19 - 14x_1 + 3x_1^2 - 14x_2 + 6x_1x_2 + 3x_2^2)] \\ & \times [30 + (2x_1 - 3x_2)^2(18 - 32x_1 + 12x_1^2 + 48x_2 - 36x_1x_2 + 27x_2^2)],\end{aligned} \tag{13}$$

which has four minima for $-2 < x_{1,2} < 2$ (Fig. 1), with the global one equal to $\mathrm{GP} = 3$ at $x = (0, -1)$. Our method treats this function as shown in Table 1 and Eqs. (15)-(17). The global minimum survives the deformation at any

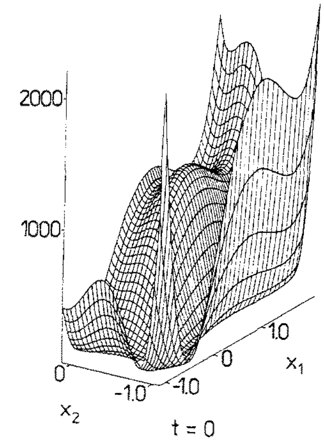

Fig. 1A. Time evolution of the Goldstein-Price function, Eq. (13). Original GP function at time $t = 0$. One can see four minima with the global one at $x = (0, -1)$. For technical reasons (a very steep ascent of the function in certain directions), the axes actually displayed correspond to a rotated coordinate system. The rotation angle is equal to arctan(2/3).

time t. Three local minima with values GP = 30, 84, 840 and basin depths 5, 15, 140, as well as the unique maximum and four saddle points disappear as the time t increases (Fig. 1) by mutual annihilation (always an extremum with a saddle point). The disappearance times for the local minima are $t = 0.0097$, 0.0618, 0.0101, respectively. The unique minimum at $t = t_0 = 1$ corresponds to $x = (0.642, -0.030)$, a point attained by minimization from any point $x = (x_1, x_2)$. The reversing procedure (Section 1) that is gradually

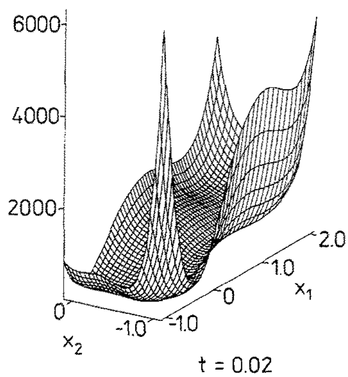

Fig. 1B. Time evolution of the Goldstein-Price function, Eq. (13). Deformed function GP(x, t) at time $t = 0.02$.

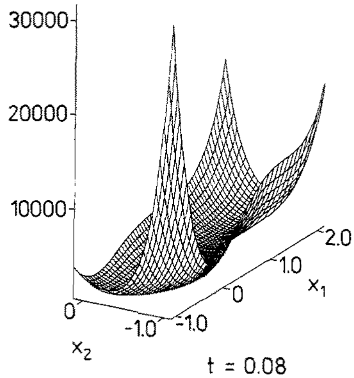

Fig. 1C. Time evolution of the Goldstein-Price function, Eq. (13). The deformed function $GP(x, t)$ at time $t = 0.08$ exhibits only a single minimum. The single minimum can be attained by a gradient-trajectory procedure from any x; then, by the reversing procedure, the global minimum can be found.

Table 1. Time evolution of the Goldstein-Price function GP leads to the global minimum independently of the starting point chosen in the procedure.

t	x_1	x_2	$GP(x, t)$
$1.000^{(a)}$	0.642	−0.030	9.003×10^5
$0.300^{(b)}$	0.469	−0.161	1.298×10^4
$0.200^{(b)}$	0.251	−0.314	4.476×10^3
$0.100^{(b)}$	−0.056	−0.532	8.031×10^2
$0.050^{(b)}$	−0.152	−0.671	2.241×10^2
$0.030^{(b)}$	−0.102	−0.796	1.161×10^2
$0.010^{(b)}$	−0.027	−0.939	2.806×10^1
$0.005^{(b)}$	−0.013	−0.970	1.293×10^1
$0.000^{(c)}$	0.000	−1.000	3.000×10^1

$^{(a)}$ For $t > t_0 = 1$, the deformed function $F(x, t)$ exhibits a single minimum at $x = (x_1, x_2, \ldots)$; the function value is given in the last column of this row.

$^{(b)}$ Selected results of the reversing procedure described in Section 1. Starting in a gradient-trajectory algorithm from the position of the minimum at $t - \Delta t$, one obtains a minimum of $F(x, t)$ at $x = (x_1, x_2, \ldots)$; the corresponding $F(x, t)$ value is displayed in the last column of this row.

$^{(c)}$ Finally, the reversing procedure leads to the global minimum at $x = (x_1, x_2, \ldots)$; the function value $f = F(x, 0)$ is shown in the last column of this row.

Table 2. Time evolution of the Hartman function H3 leads to the global minimum independently of the starting point chosen in the procedure.

t	x_1	x_2	x_3	$H3(x, t)$
$1.000^{(a)}$	0.163	0.575	0.762	−0.064
$0.400^{(b)}$	0.161	0.587	0.747	−0.180
$0.082^{(b)}$	0.138	0.623	0.741	−0.829
$0.007^{(b)}$	0.117	0.577	0.822	−2.788
$0.000^{(c)}$	0.115	0.556	0.853	−3.863

For (a), (b), (c), see Table 1.

removing the deformation, each step followed by a minimization, leads to $x = (0, -1)$, i.e., to the global minimum.

The next functions studied were H3 (in three dimensions) and H6 (in six dimensions), belonging to the Hartman's family (Ref. 7); see the Appendix, Eq. (18). The functions H3 and H6 exhibit three and two minima, respectively. These numbers are less than the number of terms (four), because of strong minima interference.

For H3, the two local minima disappear at $t = 0.0042, 0.0116$. For $t = 0$, they have the values $H3 = -3.09, -1.00$, respectively. The global minimum, having at $t = 0$ the value $H3 = -3.86$ at $x = (0.115, 0.556, 0.853)$, survives at any time t (Table 2).

For H6, the local minimum disappearance time is $t = 0.0695$. The original function value is $H6 = -3.20$. The H6 global minimum located at $x = (0.202, 0.150, 0.477, 0.275, 0.312, 0.657)$, with its value $H6 = -3.32$, does not disappear at any time t (Table 3).

The next functions studied belonged to the Shekel family (Ref. 7); see the Appendix, Eq. (21). The functions S5, S7, S10 in four dimensions contain

Table 3. Time evolution of the Hartman function H6 leads to the global minimum independently of the starting point chosen in the procedure.

t	x_1	x_2	x_3	x_4	x_5	x_6	$H6(x, t)$
$1.000^{(a)}$	0.358	0.661	0.698	0.490	0.199	0.346	−0.002
$0.400^{(b)}$	0.344	0.613	0.621	0.461	0.252	0.368	−0.015
$0.082^{(b)}$	0.246	0.303	0.530	0.297	0.311	0.626	−0.176
$0.007^{(b)}$	0.203	0.155	0.474	0.272	0.314	0.660	−2.044
$0.000^{(c)}$	0.202	0.150	0.477	0.275	0.312	0.657	−3.322

For (a), (b), (c), see Table 1.

Table 4. Time evolution of the Shekel function S5 leads to the global minimum independently of the starting point chosen in the procedure.

t	x_1	x_2	x_3	x_4	$S5(x, t)$
6.000[a]	4.693	5.658	4.693	5.658	−0.141
3.791[b]	4.712	5.698	4.712	5.698	−0.193
1.141[b]	4.455	5.132	4.455	5.132	−0.373
0.143[b]	4.004	4.015	4.004	4.015	−1.416
0.017[b]	4.000	4.001	4.000	4.001	−5.053
0.000[c]	4.000	4.000	4.000	4.000	−10.153

For (*a*), (*b*), (*c*), see Table 1.

Table 5. Time evolution of the Shekel function S7 leads to the global minimum independently of the starting point chosen in the procedure.

t	x_1	x_2	x_3	x_4	$S7(x, t)$
6.000[a]	4.355	5.773	3.982	5.400	−0.195
3.791[b]	4.398	5.656	3.986	5.245	−0.265
1.141[b]	4.438	4.806	3.774	4.142	−0.524
0.143[b]	4.069	4.082	3.940	3.951	−1.672
0.017[b]	4.004	4.004	3.997	3.998	−5.303
0.000[c]	4.001	4.001	3.999	4.000	−10.403

For (*a*), (*b*), (*c*), see Table 1.

5, 7, 10 terms and exhibit 5, 7, 10 minima, respectively. In all cases, the global minima survive (Tables 4, 5, 6), their values being $S_5 = -10.153$, $S_7 = -10.403$, $S_{10} = -10.536$.

Table 6. Time evolution of the Shekel function S10 leads to the global minimum independently of the starting point chosen in the procedure.

t	x_1	x_2	x_3	x_4	$S10(x, t)$
6.000[a]	5.226	4.545	4.930	4.249	−0.266
3.791[b]	5.265	4.464	4.931	4.130	−0.356
1.141[b]	4.897	4.371	4.368	3.842	−0.658
0.143[b]	4.087	4.071	3.958	3.941	−1.805
0.017[b]	4.005	4.004	3.998	3.997	−5.436
0.000[c]	4.001	4.001	4.000	4.000	−10.536

For (*a*), (*b*), (*c*), see Table 1.

Table 7. Time evolution of the camelback function C6 leads to one of the two (equal by symmetry) global minima independently of the starting point chosen in the procedure.

t	x_1	x_2	$C(x, t)$
$0.075^{(a)}$	0.051	−0.251	1.294
$0.030^{(b)}$	0.088	−0.574	−0.413
$0.010^{(b)}$	0.090	−0.670	−0.802
$0.000^{(c)}$	0.090	−0.713	−1.032

For (a), (b), (c), see Table 1.

Next, we considered the so-called six-hump camelback function C6 (Ref. 7)

$$C6(x_1, x_2) = 4x_1^2 - 2.1x_1^4 + (1/3)x_1^6 + x_1x_2 - 4x_2^2 + 4x_2^4, \tag{14}$$

having three conjugate pairs of minima [due to the inversion center at $x = (0, 0)$] with the values $C6 = -1.032, -0.215, 2.104$ and the corresponding basin depths 1.03, 0.76, 0.13. The two pairs of higher minima disappear at $t = 0.043$ and 0.015, respectively; see Table 7 and Fig. 2. Then, at $t = 0.086$, the two conjugate global minima and the central saddle point collapse,

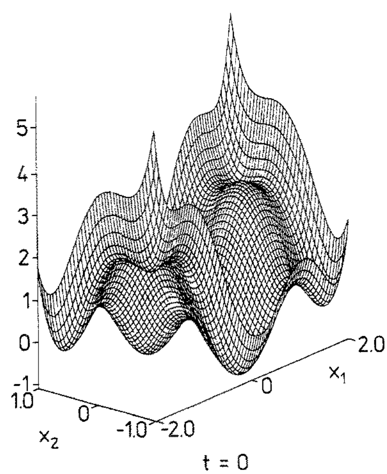

Fig. 2A. Time evolution of the six-hump camelback function C6, Eq. (14). Original C6 function at time $t = 0$. One can see three pairs of conjugate minima with the global one at $x = (0.1, -0.7)$ and $x = (-0.1, 0.7)$.

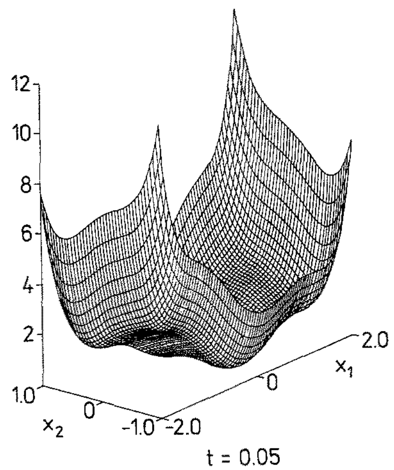

Fig. 2B. Time evolution of the six-hump camelback function C6, Eq. (14). Deformed function $C6(x, t)$ at time $t = 0.05$.

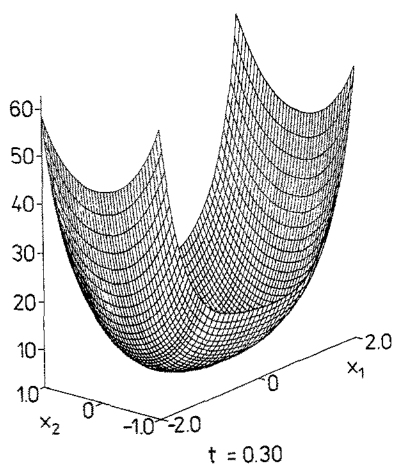

Fig. 2C. Time evolution of the six-hump camelback function C6, Eq. (14). The deformed function $C6(x, t)$ at time $t = 0.30$ exhibits only a single minimum, which resulted from a collapse of the two global minima and the central saddle point. This single minimum can be attained by a gradient-trajectory procedure from any $x = (x_1, x_2)$; then, by the reversing procedure, the global minima can be found.

Table 8. Number of function evaluations in various methods (Ref. 9) compared with that of the present method.

Method	Standard test functions							
	GP	H3	H6	S5	S7	S10	C6	G2
Törn	2499	2584	3447	3679	3606	3874	[a]	[a]
De Biase	378	732	806	620	788	1160	[a]	[a]
Price	2500	2400	7600	3800	4900	4400	[a]	[a]
Branin	[a]	[a]	[a]	5500	5020	4860	[a]	[a]
Boender	398	262	462	567	624	755	[a]	[a]
Snyman-Fatti	474	365	517	845	799	920	178	23,399
Present method	120	200	200	12,000[b]	12,000[b]	12,000[b]	120	0[c]

[a] Method not applied.
[b] Numerical integration algorithm involved.
[c] Solution obtained analytically.

yielding a single minimum at $x = (0, 0)$. The reversing procedure gives easily one of the global minima.

The global optimization of the Griewank standard test functions (Ref. 8) was considered; see the Appendix, Eq. (36). Probably, this represents a hard task for some other methods (Ref. 9), but is particularly simple in our procedure. For $n = 2$ and $d = 200$, the function G2 has some 500 minima, while $n = 10$ and $d = 4000$ result in several thousands of minima for G2

Table 9. Computation time in standard units (Ref. 9) compared with that of the present method.

Method	Standard test functions							
	GP	H3	H6	S5	S7	S10	C6	G2
Törn	4	8	16	10	13	15	[a]	[a]
De Biase	15	16	21	23	20	30	[a]	[a]
Price	3	8	46	14	20	20	[a]	[a]
Branin	[a]	[a]	[a]	9	8.5	9.5	[a]	[a]
Boender	0.5	0.7	1.7	1.3	1.7	2.5	[a]	[a]
Snyman-Fatti	0.2	0.6	1.3	1.1	1.3	2.0	0.1	100
Present method	0.04	0.3	0.5	15[b]	19[b]	26[b]	0.05	0[c]

[a] Method not applied.
[b] Numerical integration algorithm involved.
[c] Solution obtained analytically.

(Ref. 9). For large time t, the second term in Eq. (37) vanishes, while the first term represents a moving paraboloid with fixed position of its minimum at $x=0$. This position is to be found by the reversing procedure of the diffusion equation method and indeed represents the global minimum of f. Thus, in this case, one gets the result analytically without any computation.

The effectiveness of the method in the present paper has been checked by performing the test applied previously to the methods designed by other authors (Ref. 9). In Tables 8 and 9, the number of function evaluations and the computation time in standard units (Ref. 7) needed to find the global minimum has been compared for various methods. It turns out that, except for the Shekel family of functions, our method is faster than the others. This is particularly true for the Griewank functions (e.g., G2 in Tables 8 and 9), where our method finds the global minimum analytically. The Shekel family of functions leads in our method to a numerical integration (see the Appendix), and therefore to relatively long computation times.

4. Appendix A: Solution of the Diffusion Equation for Standard Test Functions

Polynomials. Consider an initial function $f:\mathbb{R}^n \to \mathbb{R}$ of the form

$$f(x)=\sum_{i_1,i_2,\ldots,i_n} a_{i_1,i_2,\ldots,i_n} x_1^{i_1} x_2^{i_2} \cdots x_n^{i_n}. \tag{15}$$

The solution of Eq. (11) with boundary condition (15) takes the form

$$\begin{aligned} F(x,t) &= \exp(t\Delta)f(x)=\prod_{j=1}^{n} \exp(t(\partial^2/\partial x_j^2))f(x) \\ &= \sum_{i_1,i_2,\ldots,i_n} a_{i_1,i_2,\ldots,i_n} \prod_{j=1}^{n} [\exp(t(\partial^2/\partial x_j^2))x_j^{i_j}] \\ &= \sum_{i_1,i_2,\ldots,i_n} a_{i_1,i_2,\ldots,i_n} W_{i_1}(x_1,t)\, W_{i_2}(x_2,t) \cdots W_{i_n}(x_n,t), \end{aligned} \tag{16}$$

where $W_n(x,t)$ is a finite sum,

$$\begin{aligned} W_n(x,t) &\equiv \exp(t(\partial^2/\partial x^2))x^n = \sum_{k=0}^{\infty} (t^k/k!)(\partial^{2k}/\partial x^{2k})x^n \\ &= \sum_{k=0}^{[n/2]} [n!/k!(n-2k)!]t^k x^{n-2k}. \end{aligned} \tag{17}$$

Hartman Functions. The objective function has the form (Ref. 7)

$$f(x)=-\sum_{i=1}^{m} c_i \exp\left[-\sum_{j=1}^{n} a_{ij}(x_j-p_{ij})^2\right]. \tag{18}$$

Applying the Fourier–Poisson formula (12) to a single Gaussian function in $\mathbb{R}^1$, $f(x)=\exp(-mx^2)$, we obtain after integration

$$F(x, t)=(1+4mt)^{-1/2}\exp(-mx^2/(1+4mt)). \tag{19}$$

Then, in $\mathbb{R}^n$, we have for the full function

$$F(x, t)=-\sum_{i=1}^{m} c_i \prod_{j=1}^{n}\left\{(1+4a_{ij}t)^{-1/2}\exp[-a_{ij}(x_j-p_{ij})^2/(1+4a_{ij}t)]\right\}. \tag{20}$$

Shekel Functions. The objective function is of the form (Ref. 7)

$$f(x)=-\sum_{i=1}^{m}\left[c_i+\sum_{j=1}^{n}(x_j-a_{ij})^2\right]^{-1}. \tag{21}$$

Consider a single term centered at $(0, 0, \ldots, 0)$,

$$\phi(x, c)=(c+\|x\|^2)^{-1}. \tag{22}$$

The solution for ϕ can be expressed according to the Fourier–Poisson formula as

$$\Phi(x, t, c)=(2\sqrt{\pi t})^{-n}I(x, 1/4t, c), \tag{23}$$

$$I(x, \lambda, c)\equiv\int_{\mathbb{R}^n}[\exp(-\lambda\|y-x\|^2)/(c+\|y\|^2)]\,dy. \tag{24}$$

The function $I(x, \lambda, c)$ is spherically symmetric in x; i.e., if $\|a\|=\|b\|$, then $I(a, \lambda, c)=I(b, \lambda, c)$. This means that we can restrict our study to the function

$$J(\chi, \lambda, c)\equiv I\left(\begin{bmatrix}\chi\\0\\0\\\vdots\\0\end{bmatrix}, \lambda, c\right). \tag{25}$$

For any vector a,

$$I(a, \lambda, c)=J(\|a\|, \lambda, c).$$

Let us use spherical coordinates of the form

$$y_1=r\cos\omega_1, \tag{26a}$$

$$y_2=r\sin\omega_1\cos\omega_2, \tag{26b}$$

$$\cdots$$

$$y_{n-1}=r\sin\omega_1\sin\omega_2\cdots\sin\omega_{n-2}\cos\omega_{n-1}, \tag{26c}$$

$$y_n=r\sin\omega_1\sin\omega_2\cdots\sin\omega_{n-2}\sin\omega_{n-1}, \tag{26d}$$

for $0 \leq \omega_j \leq \pi$, $j = 1, \ldots, n-2$ and $0 \leq \omega_{n-1} \leq 2\pi$, with Jacobian

$$D = r^{n-1} \sin^{n-2} \omega_1 \cdots \sin \omega_{n-2}$$

and

$$J(\chi, \lambda, c) = 2\pi \int_0^{\pi} \sin \omega_{n-2} \, d\omega_{n-2} \cdots \int_0^{\pi} \sin^{n-3} \omega_2 \, d\omega_2 \times \int_0^{\infty} \int_0^{\pi} r^{n-1}(c+r^2)^{-1} \sin^{n-2} \omega_1 \times \exp[-\lambda(r^2 - 2r\chi \cos \omega_1 + \chi^2)] \, d\omega_1 \, dr. \tag{27}$$

In the Shekel functions, $n = 4$. After replacing ω_1 by ω, one has

$$\Phi(x, t, c) = (4\lambda^2/\pi) \int_0^{\infty} \int_0^{\pi} [r^3/(c+r^2)] \sin^2 \omega \times \exp[-\lambda(r^2 - 2\chi r \cos \omega + \chi^2)] \, d\omega \, dr. \tag{28}$$

Taking into account that

$$\int_0^{\pi} \sin^2 \omega \exp(\alpha \cos \omega) \, d\omega = (\pi/\alpha) I_1(\alpha), \tag{29}$$

where I_1 is the first-order modified Bessel function (Ref. 10), one has

$$\Phi(x, t, c) = (2\lambda/\chi) \exp(-\lambda\chi^2) \times \int_0^{\infty} [r^2/(c+r^2)] \exp(-\lambda r^2) I_1(\gamma r) \, dr, \tag{30}$$

with $\gamma \equiv 2\lambda\chi$. The integral on the right-hand side of (30) can be transformed as follows:

$$\begin{aligned}
&\int_0^{\infty} [r^2/(c+r^2)] \exp(-\lambda r^2) I_1(\gamma r) \, dr \\
&= \int_0^{\infty} \exp(-\lambda r^2) I_1(\gamma r) \, dr - c \int_0^{\infty} [1/(c+r^2)] \exp(-\lambda r^2) I_1(\gamma r) \, dr \\
&= \int_0^{\infty} \exp(-\lambda r^2) I_1(\gamma r) \, dr \\
&\quad - c \exp(\lambda c) \int_0^{\infty} \int_{\lambda}^{\infty} \exp[-u(c+r^2)] \, du \, I_1(\gamma r) \, dr \\
&= \int_0^{\infty} \exp(-\lambda r^2) I_1(\gamma r) \, dr \\
&\quad - c \exp(\lambda c) \int_{\lambda}^{\infty} \exp(-uc) \int_0^{\infty} \exp(-ur^2) I_1(\gamma r) \, dr \, du.
\end{aligned} \tag{31}$$

Noting that (Ref. 11)

$$\int_0^\infty \exp(-pr^2) I_1(\gamma r)\, dr = \gamma^{-1}[\exp(\gamma^2/4p) - 1], \tag{32}$$

formula (31) can be simplified to the form

$$\begin{aligned}&\int_0^\infty [r^2/(c+r^2)] \exp(-\lambda r^2) I_1(\gamma r)\, dr \\ &= \gamma^{-1}\left[\exp(\gamma^2/4\lambda) - c\exp(\lambda c)\int_\lambda^\infty \exp(-cu + \gamma^2/4u)\, du\right].\end{aligned} \tag{33}$$

Thus,

$$\Phi(x, t, c) = \chi^{-2}\left[1 - c\exp[\lambda(c-\chi^2)]\int_\lambda^\infty \exp(-cu + \lambda^2\chi^2/u)\, du\right]; \tag{34}$$

after the substitution $cu = 1/v$, one gets finally

$$\Phi(x, t, c) = c\lambda^2 \exp[\lambda(c-\chi^2)]\int_0^{1/\lambda c} \exp(-v^{-1} + c\lambda^2\chi^2 v)\, dv. \tag{35}$$

The solution for the full function, Eq. (21), is

$$F(x, t) = -\sum_{i=1}^m \Phi(x - a_i, t, c_i),$$

where

$$a_i = \begin{bmatrix} a_{i1} \\ a_{i2} \\ \vdots \\ a_{in} \end{bmatrix}.$$

Griewank Functions. The objective function is of the form

$$f(x) = d^{-1}\|x\|^2 - \prod_{j=1}^n \cos(x_j/\sqrt{j}) + 1, \qquad d > 0. \tag{36}$$

In our method, the solution of Eq. (11) with boundary condition (36) takes the form [see Eqs. (3)–(8)]

$$F(x, t) = \sum_{j=1}^n d^{-1}(x_j^2 + 2t) - \prod_{j=1}^n \exp(-t/j)\cos(x_j/\sqrt{j}) + 1. \tag{37}$$

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